How Much of DeFi Liquidity Is Higher-Order? Survivorship, LP Representation, and a Robust Null

Adam Levine
Department of Economics, Hofstra University
adalevine08@gmail.com

July 23, 2026

Abstract

A multi-asset automated-market-maker pool (e.g. Curve's 3pool $\{\mathrm{USDC}, \mathrm{USDT}, \mathrm{DAI}\}$) encodes a genuinely three-way relationship that a pairwise graph discards, motivating a persistent-homology treatment of the *nerve complex* of pool co-membership. We ask how much of the loop structure of Ethereum stablecoin liquidity is genuinely higher-order, and how stable that quantity is. The answer is that **it is not stable at all**: the measured higher-order fraction swings from 0.31 to 0.93 across two defensible and rarely-scrutinised modeling choices. The larger driver is **LP-token representation** (keeping an LP token such as 3CRV as its own vertex vs. resolving it to base assets, which collapses metapools toward a cone and roughly doubles the fraction); the second is **survivorship**. On the latter we make a methodological contribution: the standard registry-based historical reconstruction, used across DeFi network research, is severely survivor-biased — reconstructing from pools that survive to today captures only $\sim$10 % of the loop structure — and it can be *de-censored* using Internet Archive snapshots of the registry endpoint, which we do over 36 snapshots (Oct 2022–May 2025). De-censoring also revives the LP-resolution fork that a survivor-only view cannot even run, because the relevant metapools have all delisted. Against this fragility of the descriptive metric, one result is robust to every choice we vary — survivorship, LP representation, and sampling cadence: through the March 2023 USDC depeg the reconstructed structure is **inert**, changing less than routine month-to-month drift. We read this economically (stablecoins are common-numéraire substitutes, so their liquidity scaffold is shallow and price-stress-invariant) and draw the methodological lesson: report the higher-order content of a DeFi network as a sensitivity range over survivorship and representation, never as a point estimate. Code and data are fully reproducible.

1 Introduction

Classical financial-contagion models are pairwise: institutions are nodes, bilateral exposures are edges (Eisenberg & Noe, 2001). A multi-asset AMM pool is not — Curve's 3pool binds {USDC, USDT, DAI} into one contract, and a graph records only the three pairwise projections, discarding the joint membership. Higher-order network methods (hypergraphs, simplicial complexes; Bick et al., 2023) retain exactly that information, and a nascent literature argues they matter for systemic risk (Forte, 2026). The decisive reason to test this in decentralised finance rather than traditional finance is **observability**: pool compositions and reserves are public and machine-readable on-chain, removing the confidentiality barrier that has always constrained structural contagion research.

We make "how much is higher-order?" a measurement. For each snapshot we build the nerve complex of pool co-membership and compute the *skeleton-vs-nerve decomposition*: the cycle rank of the 1-skeleton ($E - V + B_0$, the loops a graph sees) vs. the Betti-1 number B_1 of the nerve (loops surviving once multi-asset pools fill their faces). Their ratio is the fraction of loop structure that is genuinely higher-order.

The central finding is that this fraction is **not a property of the network but of the analyst's choices**. Two knobs that any such study must set, and that are rarely reported as sensitivities, move it from a clear minority to an overwhelming majority:

1. **LP-token representation.** A Curve metapool reads nominally as $\{X, \text{3CRV}\}$. Resolving 3CRV into its underlying $\{\text{USDC}, \text{USDT}, \text{DAI}\}$ makes every such pool a simplex containing the 3pool triangle as a face, pushing the complex toward a cone (contractible, higher-order by construction); keeping 3CRV as its own vertex does not. This choice is the *larger* driver.
2. **Survivorship.** The standard historical reconstruction filters *today's* registry back in time, so it silently conditions on pools that survived — disproportionately the large multi-asset pools. This inflates the higher-order fraction and, we show, is fixable.

Our contributions:

- **A de-censoring method for DeFi registry survivorship** (§3): Internet-Archive snapshots of the registry recover the full historical universe, delisted pools included. Survivor-only reconstruction captures ∼1/10 of the loops (§5.1); the method applies to any registry-based DeFi network study.
- **A sensitivity decomposition of the higher-order fraction** (§5.2): it spans 0.31–0.93 across representation × survivorship; we separate the two contributions and show even the *sign of the time-trend* is convention-dependent. De-censoring revives the otherwise-untestable LP-resolution fork.
- **A convention-robust null** (§5.3): the reconstructed structure is inert through the USDC depeg under every choice we vary, and this is the one representation- and sample-independent structural statement the data supports.

A rigorous negative on the dynamics, and a demonstration that the descriptive metric is an artifact, together map where higher-order topology does and does not add value over pairwise methods — which we did not know in advance.

2 Related work

Higher-order finance is early-stage. Forte (2026) give the first hypergraph analysis of bank–firm credit networks (Central Bank of Argentina registry, adjusted H-eigenvector centrality) — TradFi credit and centrality, not persistent homology and not DeFi. **Return-space TDA** is by contrast saturated: persistent homology of return point clouds covers equities (Gidea & Katz, 2018), FX, and crypto (de Favereau de Jeneret & Diamantis, 2025); we keep this lineage strictly separate, as our complex is built from *structure* (co-membership), not return correlation. **Graph analysis of DeFi** exists — Yan & Tessone (2025) model Uniswap as a token-node/pool-edge network — and is exactly the pairwise baseline our decomposition sits atop; Aufiero et al. (2025) survey the surrounding systemic-risk landscape. To our knowledge the persistent-homology / nerve treatment of DeFi liquidity structure is previously unoccupied, and, more to the point, the survivorship and representation sensitivities we quantify are not addressed in registry-based DeFi network work, which typically reconstructs history from the current registry without either control.

3 Data: the registry, and de-censoring its survivorship

We use two free, unauthenticated DeFiLlama endpoints: the pool registry (`yields.llama.fi/pools`: project, chain, symbol, `underlyingTokens`, current TVL, stablecoin flag) and per-pool daily TVL history (`yields.llama.fi/chart/{pool}`). The universe is Ethereum, stablecoin-flagged pools with 2–8 underlying tokens.

The standard reconstruction, and its hidden conditioning. Because a pool's token set is fixed at deployment, the usual historical complex at date t is today's registry filtered to pools with TVL above threshold at t. This silently conditions on survival: pools delisted before today are invisible. It has been treated (including in two independent reviews of an earlier draft of this work) as an *unrecoverable* ceiling.

De-censoring. It is not unrecoverable. The Wayback Machine archived the registry endpoint roughly weekly from October 2022; each snapshot is the *full* universe at that date — delisted pools included — with the identical schema. We fetch 36 archived snapshots (Oct 2022–May 2025, monthly with a weekly cluster around the depeg), rebuild the complex on the full historical universe, and, per snapshot, also restrict to pools surviving to today, isolating the survivorship effect at fixed representation. True survivorship rises from 24 % to 41 % over the period (mechanically — recent pools are likelier to persist); at the depeg only 66 of 274 universe pools survive, and the 208 invisible pools were 35 % of universe TVL (FRAX-3CRV \$474M, MIM/OUSD/TUSD/LUSD-3CRV, . . .). Terra (May 2022) predates the archive and is not de-censorable; snapshots are weekly crawl dates, not daily.

4 Method

Complex and filtration. Vertices are tokens (by contract address); a pool with token set σ contributes the filled simplex on σ (downward closure). We filter by each pool's *share* of universe TVL at date t, entering strong pools first at $f = -\log_{10}(\text{share})$ — a level-invariant weighting, so a result cannot be the mechanical TVL drawdown of a depeg in disguise. We track B_0, essential B_1, B_2 ($\equiv 0$ throughout), the 1-skeleton cycle rank $E - V + B_0$, and the higher-order fraction $\text{HO} = (E - V + B_0 - B_1)/(E - V + B_0)$. Topology code is validated on hand-checkable toy complexes (a hollow triangle has one loop, a filled one none, a tetrahedron boundary has $B_2 = 1$).

The two modeling axes. We cross **sample** (survivor-only vs. de-censored/full) with **LP representation**: *lp_vertex* keeps LP tokens as their own vertices (the archive's native form), *resolved* expands the 3CRV LP token (identified as `0x6c3f...e490`, present in ∼68 metapools per snapshot in the archive) into $\{\text{DAI}, \text{USDC}, \text{USDT}\}$. Today's API base-resolves, so the survivor reconstruction is implicitly the *resolved* fork; the archive's native form makes *lp_vertex* — and thus the comparison — available for the first time.

Event test. With snapshots at weekly cadence we test the depeg non-parametrically: is the pre→post-depeg change (2023-03-07 → 03-16) large relative to the distribution of all consecutive-snapshot changes across the series? We run it under both representations. (An earlier daily survivor-only analysis using a placebo-window permutation test over 632 daily complexes reaches the same null; we report the de-censored version as the primary evidence.)

snapshot	lp_vertex, full	lp_vertex, survivor	resolved, full	resolved, survivor
2022-10	0.31	0.73	0.80	0.93
2023-03	0.37	0.73	0.75	0.93
2023-06	0.40	0.67	0.68	0.92
2023-11	0.30	0.65	0.66	0.87
2024-05	0.32	0.52	0.54	0.77
2024-12	0.39	0.48	0.54	0.68
2025-05	0.40	0.39	0.50	0.60

Table 1: Higher-order fraction by sample × LP representation. The paper-standard corner (resolved, survivor) reads a declining majority; the de-censored, composability- preserving corner (lp_vertex, full) reads a flat minority. Same data.

5 Results

5.1 Survivor reconstruction sees ∼1/10 of the loop structure

At fixed representation (lp_vertex), essential B_1 is **40–58 on the full universe vs. 4–23 on survivors**, at every snapshot; the true universe is 3–4× the survivor universe (234–430 vs. 63–178 pools). The object a survivor-only study analyses is a small, biased slice — and the bias is toward exactly the large multi-asset pools that dominate the higher-order fraction.

5.2 The higher-order fraction is a modeling artifact

Table 1 crosses the two axes. The higher-order fraction spans **0.31–0.93**. Reading it two ways:

- **Representation is the larger driver.** Resolving 3CRV collapses ∼68 metapools onto a shared base triangle (a partial cone): across 36 snapshots it removes ∼28 % of loops (essential B_1 46.8 → 33.8) yet nearly *doubles* the higher-order fraction (0.36 → 0.67). At the depeg, full-universe HO is 0.37 (lp_vertex) vs. 0.75 (resolved).
- **Survivorship amplifies in the same direction.** Holding representation = resolved (the paper-standard convention), full→survivor pushes HO 0.75 → 0.93 at the depeg, because survivors are the big multi-asset pools.
- **Even the trend is convention-dependent** (Fig. 1). Under resolved, the de-censored fraction genuinely *declines* 0.80 → 0.50 as metapools delist; under lp_vertex it is *flat* at 0.36 ± 0.04. The survivor fraction declines under both — but that is the survivor sample converging to the full-universe value as survivorship rises (corr(survivor HO, survivorship) = −0.92 for lp_vertex), not a change in the network.

So the paper-standard "a majority of loops are higher-order, declining over 2022–23" sits at the high-inflation corner; the de-censored, composability-preserving reading is a flat minority. Neither convention is uniquely correct (resolving reflects economic exposure to base assets; lp_vertex reflects composability structure). **The honest answer to "how much is higher-order?" is the range in Table 1, not a number.**

5.3 The null is the robust part

Against all of that fragility, the dynamics are stable. Table 2 runs the de-censored event test: the change across the depeg is *smaller* than typical between-snapshot drift for every observable under

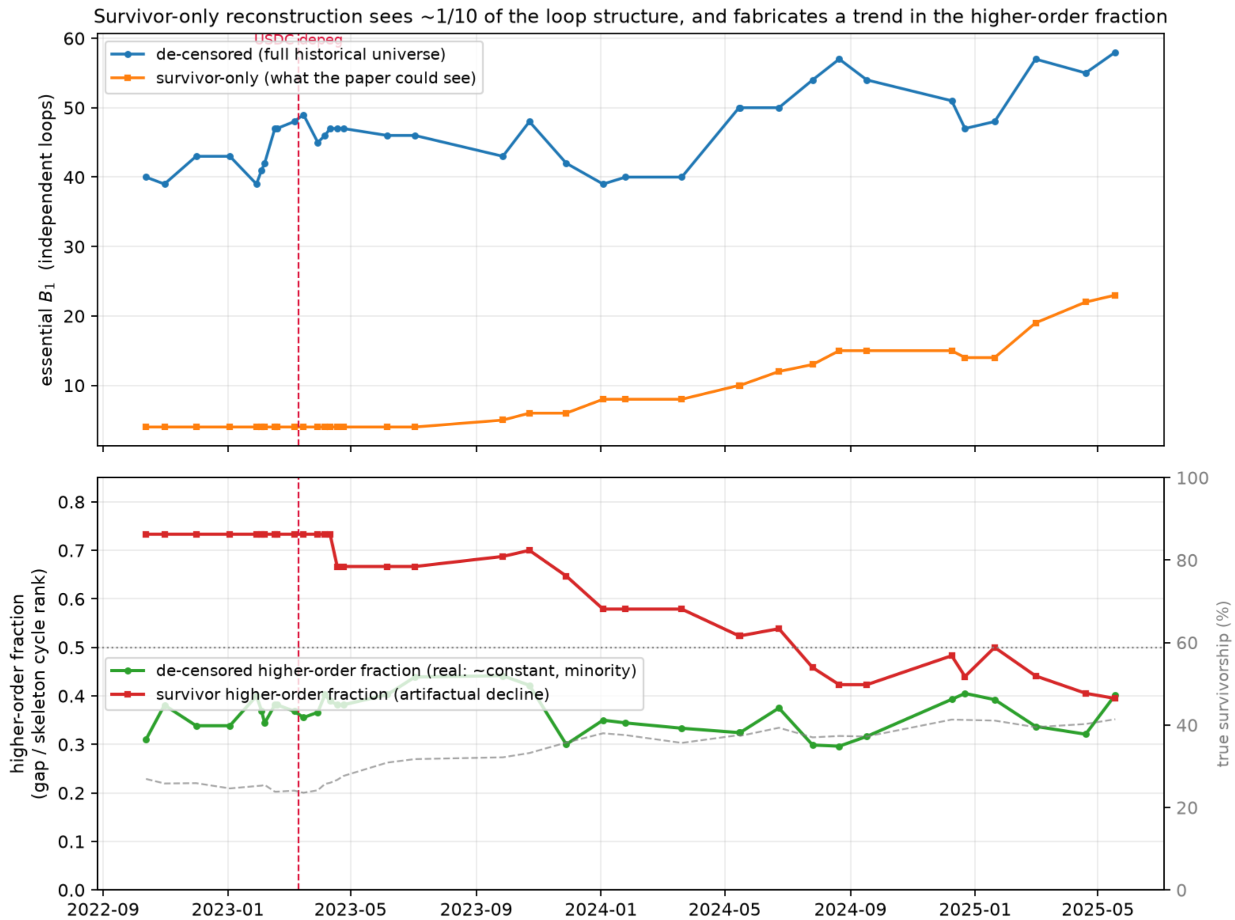

Figure 1: Survivor-only reconstruction (orange/red) sees ∼1/10 of the loops (top) and manufactures a declining higher-order trend (bottom) that the de-censored full universe (blue/green) does not show; the survivor decline tracks rising survivorship (grey dashed), which explains it. Vertical line: the USDC depeg.

both representations — essential B_1 under resolved does not move at all.

Unlike the descriptive fraction, the null is robust to every choice we can vary — survivorship, LP representation, and cadence. It directly refutes the natural objection that the survivor view looks rigid only because it omits the pools that broke: restoring all 208 dead pools, the structure is still inert.

6 Discussion

The fragility and the null resolve under one economic reading. Stablecoins are near-perfect substitutes pegged to a common numéraire, so the liquidity network is a hub-and-spoke around "dollarness" — mostly the 3pool and shallow wrappers — rather than a rich web of distinct multi-way dependencies. Two consequences follow: (i) *how much* of that shallow scaffold counts as "higher-order" is genuinely under-determined, because whether a metapool's dependence on the 3pool is one relationship (an LP vertex) or three (resolved base assets) is a modeling stance, not a fact — hence the 0.31–0.93 range; and (ii) a depeg is prices and flows moving *within* a fixed scaffold, not the scaffold re-wiring — hence the null.

observable	pre→post depeg $\lvert\Delta\rvert$	percentile of $\lvert$consecutive $\Delta\rvert$	verdict
essential B_1, lp_vertex	1	23rd	inert
gap, lp_vertex	1	14th	inert
essential B_1, resolved	0	0th	inert
gap, resolved	1	17th	inert

Table 2: De-censored event test, full universe (2023-03-07 → 03-16). The largest stablecoin depeg on record moves the structure less than routine month-to-month drift, under both LP conventions.

The transferable lesson is methodological. Registry-based reconstruction is the default in empirical DeFi network research, and it silently conditions on survival; we show that conditioning discards ∼90% of the loop structure and, combined with an unremarked representation choice, can flip a headline descriptive claim. Both are cheap to address: de-censor via the Internet Archive, and report the representation sensitivity. Where structural higher-order TDA might still earn its keep is where these caveats bite less — non-substitute universes (mixed collateral, cross-asset lending) with heterogeneous multi-way dependence, and slow regime changes that actually re-wire membership.

7 Limitations

- **Two defensible conventions, no tie-breaker.** We report the range rather than adjudicate; a reader with a specific economic question (exposure vs. composability) should pick the matching convention and its number.
- **Terra (May 2022) is not de-censorable** — it predates the archive. Snapshots are weekly, so the de-censored event test is coarse (we corroborate with a daily survivor-only placebo permutation reaching the same null).
- **Ethereum stablecoins only.** The economic reading (common-numéraire substitutes) is exactly why the scaffold is shallow; other universes may differ, which is the point.
- **Cross-era representation.** The archive is natively lp_vertex and today's API resolved; we run both explicitly rather than mixing them.

8 Conclusion

Measured rather than asserted, the higher-order structure of Ethereum stablecoin liquidity has no convention-free size: the higher-order fraction ranges 0.31–0.93 across two modeling choices — LP-token representation (the larger) and survivorship — that registry- based DeFi network studies routinely leave unexamined, and even its time-trend flips sign between conventions. Survivor-only reconstruction, the field default, sees roughly a tenth of the loop structure; Internet-Archive de-censoring recovers the rest and revives an otherwise-untestable representation fork. The one convention-independent structural result is a null: through the largest stablecoin depeg on record the reconstructed structure is inert, moving less than routine drift. Report higher-order DeFi structure as a sensitivity range, de-censor the registry, and treat the robust dynamics — not the fragile levels — as the finding.

Reproducibility

All code, cached observables, archived snapshots, and figures are at `https://github.com/adamLEVi16/topology-engine` (`defi_topology/`). The pipeline runs from the free DeFiLlama API and the Internet Archive with no authentication; toy-complex tests pin the topology, and every number is regenerated by a single script per result (`decensor.py` for the de-censored series and the 2×2).

References

Bick, C., Gross, E., Harrington, H. A., Schaub, M. T. (2023). What are higher-order networks? *SIAM Review* 65(3).

Eisenberg, L., Noe, T. H. (2001). Systemic risk in financial systems. *Management Science* 47(2).

Forte, F. D. (2026). It takes two to tango, but more to assess systemic risk: credit networks through the lens of hypergraphs. arXiv:2607.10943.

Gidea, M., Katz, Y. (2018). Topological data analysis of financial time series: landscapes of crashes. *Physica A* 491.

de Favereau de Jeneret, P., Diamantis, I. (2025). Topology of currencies: persistent homology for FX co-movements: a comparative clustering study. arXiv:2510.19306.

Yan, T., Tessone, C. J. (2025). Network analysis of Uniswap: centralization and fragility in the decentralized exchange market. arXiv:2503.07834.

Aufiero, S., Bartolucci, S., Caccioli, F., Vivo, P. (2025). Mapping microscopic and systemic risks in TradFi and DeFi: a literature review. arXiv:2508.12007.